Student Data: Hardhat Commands and Code

Nov 4, 2026 · @Shreyash

Step 1: Install Node.js and npm

Skip this step if Node.js is already installed.

```
cd ~
sudo apt update
sudo apt install -y nodejs npm
hash -r
node -v
npm -v
```

Step 2: Create the project

Use Option A or Option B, not both.

Option A: Without copying

```
cd ~
mkdir Student_Data
cd Student_Data
mkdir contracts scripts
npm init -y
npm install hardhat@2 @nomicfoundation/hardhat-ethers@3 ethers@6 solc@0.8.24
ls
```

Create the config:

```
code hardhat.config.js
```

```
require("@nomicfoundation/hardhat-ethers");

module.exports = {
  solidity: "0.8.24"
};
```

Option B: Copying from Bank_Account

```
cd ~
mkdir Student_Data
cd Student_Data
mkdir contracts scripts
npm init -y
cp -r ~/Bank_Account/node_modules .
cp ~/Bank_Account/hardhat.config.js .
ls
```

Step 3: Create contracts/StudentData.sol

```
code contracts/StudentData.sol
```

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.24;

contract StudentData {
    struct Student {
        uint256 rollNo;
        string name;
        string course;
    }

    Student[] private students;

    event StudentAdded(uint256 rollNo, string name, string course);
    event EtherReceived(address sender, uint256 amount);

    function addStudent(uint256 rollNo, string memory name, string memory
course) public {
        students.push(Student(rollNo, name, course));
        emit StudentAdded(rollNo, name, course);
    }

    function getStudent(uint256 index) public view returns (uint256, string
memory, string memory) {
        require(index < students.length, "Invalid student index");
        Student memory s = students[index];
        return (s.rollNo, s.name, s.course);
```

```
    }

    function studentCount() public view returns (uint256) {
        return students.length;
    }

    fallback() external payable {
        emit EtherReceived(msg.sender, msg.value);
    }
}
```

Step 4: Compile

```
npx hardhat compile
```

```
Compiled 1 Solidity file successfully (evm target: paris).
```

Step 5: Create scripts/run.js

```
code scripts/run.js
```

```
const { ethers } = require("hardhat");

async function main() {
  const school = await ethers.deployContract("StudentData");

  await (await school.addStudent(1, "Student A", "Computer")).wait();
  console.log("Student:", await school.getStudent(0));
  console.log("Total students:", await school.studentCount());

  const [signer] = await ethers.getSigners();
  await (await signer.sendTransaction({ to: school.target, value: 100
})).wait();
  console.log("Contract balance:", await
ethers.provider.getBalance(school.target));
}

main();
```

Step 6: Execute

```
npx hardhat run scripts/run.js
```

```
Student: Result(3) [ 1n, 'Student A', 'Computer' ]
Total students: 1n
Contract balance: 100n
```